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JEE MAIN · QUESTION PAPER

JEE Main 2025 (Online) 22nd January Evening Shift

JEE Main Question Paper with Answer Key

22 Jan 2025	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
Mathematics	25 questions
Physics	25 questions
Answer key	at the end

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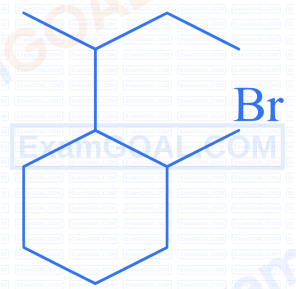
Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

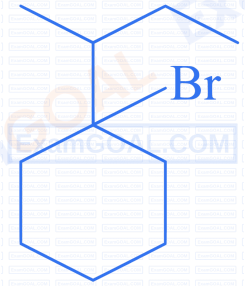
Open solution

When sec-butylcyclohexane reacts with bromine in the presence of sunlight, the major product is :

(A)



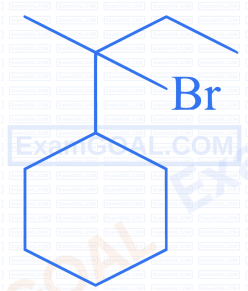
(B)



(C)



(D)



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Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

Statement (II) : The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

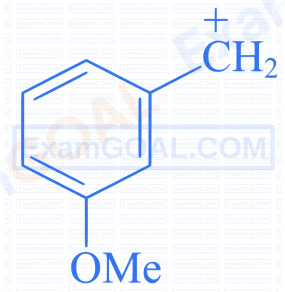
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Q3. MCQ (Single Correct Answer)

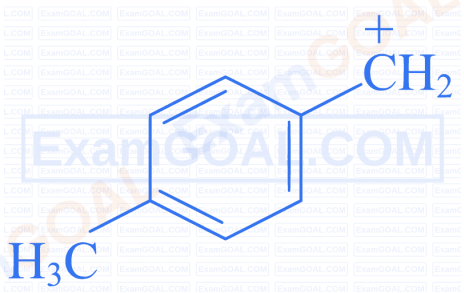
Open solution

The most stable carbocation from the following is :

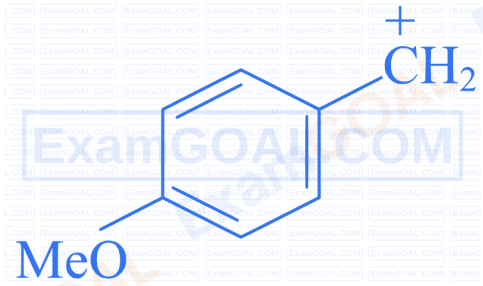
(A)



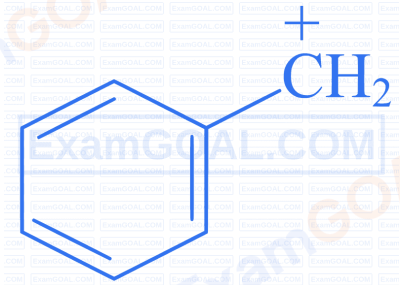
(B)



(C)



(D)



Q4. MCQ (Single Correct Answer)

Open solution

The correct order of the following complexes in terms of their crystal field stabilization energies is :

(A) $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+}$

(B) $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+}$

(C) $[Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_4]^{2+} < [Co(en)_3]^{3+}$

(D) $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_4]^{2+}$

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Q5. MCQ (Single Correct Answer)

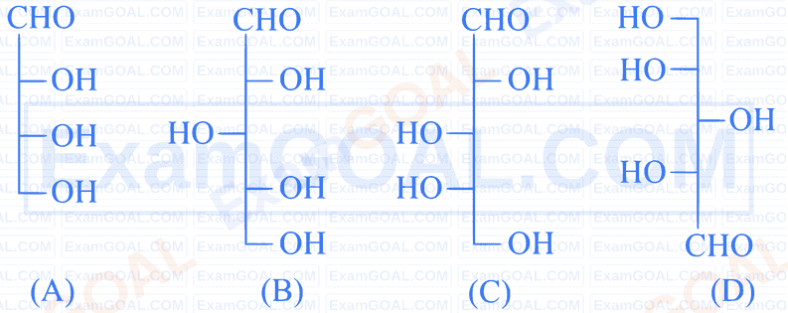
The maximum covalency of a non-metallic group 15 element ' E ' with weakest $E - E$ bond is :

(A) 5 (B) 3 (C) 4 (D) 6

Open solution

Q6. MCQ (Single Correct Answer)

Identify the number of structure/s from the following which can be correlated to D-glyceraldehyde.



(A) three (B) two (C) one (D) four

Open solution

Q7. MCQ (Single Correct Answer)

Given below are two statements :

Statement (I) : Nitrogen, sulphur, halogen and phosphorus present in an organic compound are detected by Lassaigne's Test.

Statement (II) : The elements present in the compound are converted from covalent form into ionic form by fusing the compound with Magnesium in Lassaigne's test.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

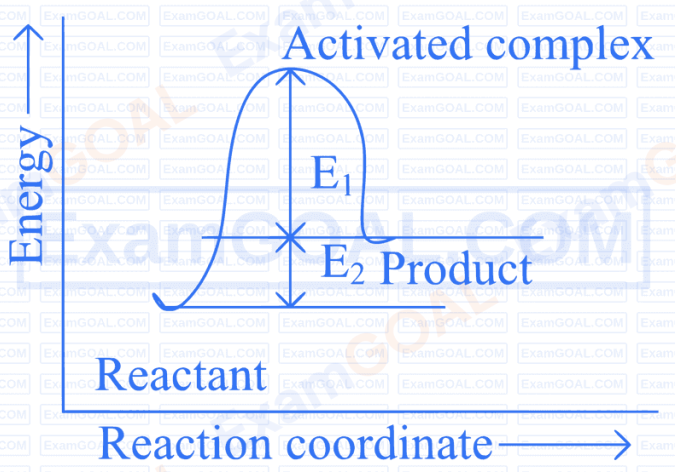
Open solution

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Q8. MCQ (Single Correct Answer)

Open solution

Consider the given figure and choose the correct option :



(A) Activation energy of both forward and backward reaction is $E_1 + E_2$ and reactant is more stable than product.

(B) Activation energy of forward reaction is $E_1 + E_2$ and product is more stable than reactant.

(C) Activation energy of backward reaction is E_1 and product is more stable than reactant.

(D) Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant.

Q9. MCQ (Single Correct Answer)

Open solution

Arrange the following compounds in increasing order of their dipole moment :

$\mathrm{HBr}, \mathrm{H_2\,S}, \mathrm{NF_3}$ and $\mathrm{CHCl_3}$

(A) $\mathrm{HBr} < \mathrm{H_2\,S} < \mathrm{NF_3} < \mathrm{CHCl_3}$

(B) $\mathrm{H_2\,S} < \mathrm{HBr} < \mathrm{NF_3} < \mathrm{CHCl_3}$

(C) $\mathrm{NF_3} < \mathrm{HBr} < \mathrm{H_2\,S} < \mathrm{CHCl_3}$

(D) $\mathrm{CHCl_3} < \mathrm{NF_3} < \mathrm{HBr} < \mathrm{H_2S}$

Q10. MCQ (Single Correct Answer)

Open solution

Density of 3 M NaCl solution is $1.25\ \mathrm{g/mL}$. The molality of the solution is :

(A) 2.79 m

(B) 2 m

(C) 1.79 m

(D) 3 m

Q11. MCQ (Single Correct Answer)

Open solution

The molar solubility(s) of zirconium phosphate with molecular formula $\left(\mathrm{Zr}^{4+}\right)_3\left(\mathrm{PO}_4^{3-}\right)_4$ is given by relation :

(A) $\left(\frac{\mathrm{K_{sp}}}{5348}\right)^{\frac{1}{6}}$

(B) $\left(\frac{\mathrm{K_{sp}}}{8435}\right)^{\frac{1}{7}}$

(C) $\left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$

(D) $\left(\frac{\mathrm{K_{sp}}}{9612}\right)^{\frac{1}{3}}$

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Q12. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I (Partial Derivatives)		List - II (Thermodynamic Quantity)
(A)	$\left(\frac{\partial G}{\partial T}\right)_P$	(I)	Cp
(B)	$\left(\frac{\partial H}{\partial T}\right)_P$	(II)	$-S$
(C)	$\left(\frac{\partial G}{\partial P}\right)_T$	(III)	Cv
(D)	$\left(\frac{\partial U}{\partial T}\right)_V$	(IV)	V

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(D) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Q13. MCQ (Single Correct Answer)

Open solution

Identify the homoleptic complex(es) that is/are low spin.

(A) $[Fe(CN)_5NO]^{2-}$

(B) $[CoF_6]^{3-}$

(C) $[Fe(CN)_6]^{4-}$

(D) $[Co(NH_3)_6]^{3+}$

(E) $[Cr(H_2O)_6]^{2+}$

Choose the correct answer from the options given below :

(A) (C) and (D) only

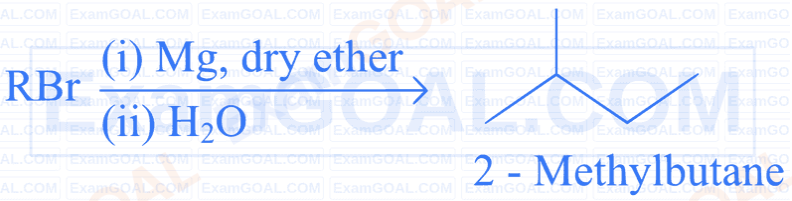
(B) (C) only

(C) (A) and (C) only

(D) (B) and (E) only

Q14. MCQ (Single Correct Answer)

Open solution



The maximum number of RBr producing 2-methylbutane by above sequence of reactions is _____ . (Consider the structural isomers only)

(A) 5

(B) 3

(C) 4

(D) 1

Q15. MCQ (Single Correct Answer)

Open solution

The alkane from below having two secondary hydrogens is :

(A) 2,2,4,5-Tetramethylheptane

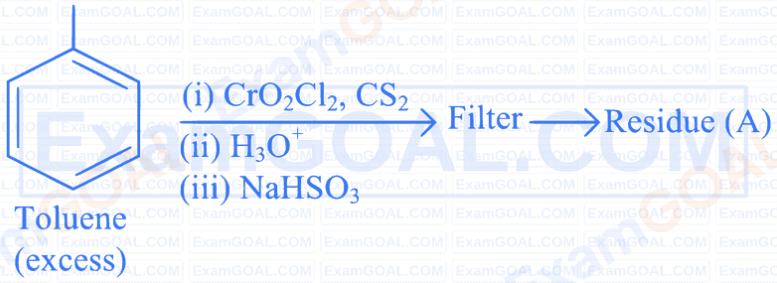
(B) 2,2,4,4-Tetramethylhexane

(C) 4-Ethyl-3,4-dimethyloctane

(D) 2,2,3,3-Tetramethylpentane

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Q16. MCQ (Single Correct Answer)

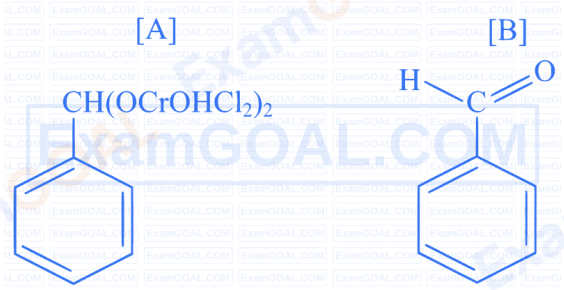


Open solution

Residue (A) + HCl (dil) $\rightarrow$ Compound (B)

Structure of residue (A) and compound (B) formed respectively is :

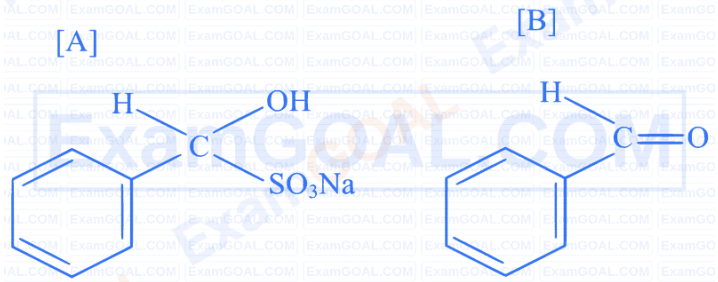
(A)



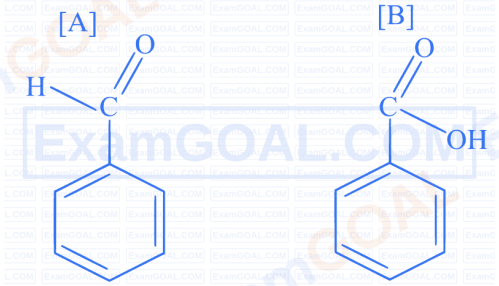
(B)



(C)



(D)



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Q17. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : An element in the extreme left of the periodic table forms acidic oxides.

Statement (II) : Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q18. MCQ (Single Correct Answer)

Open solution

The species which does not undergo disproportionation reaction is :

(A) ClO_4^- (B) ClO_3^- (C) ClO^- (D) ClO_2^-

Q19. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition.

Statement (II) : $2p_x$ and $2p_y$ are degenerate orbitals.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

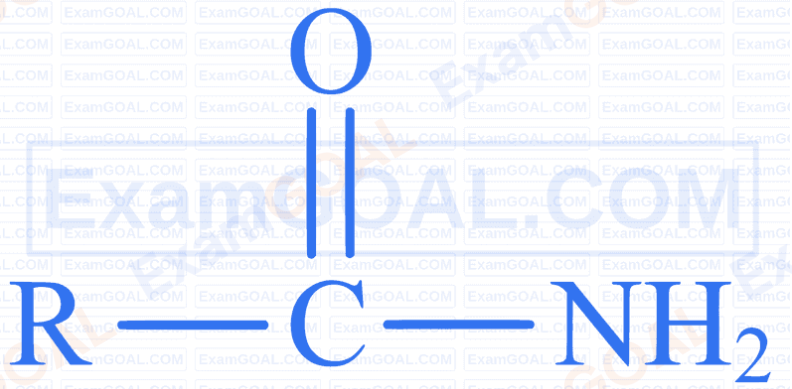
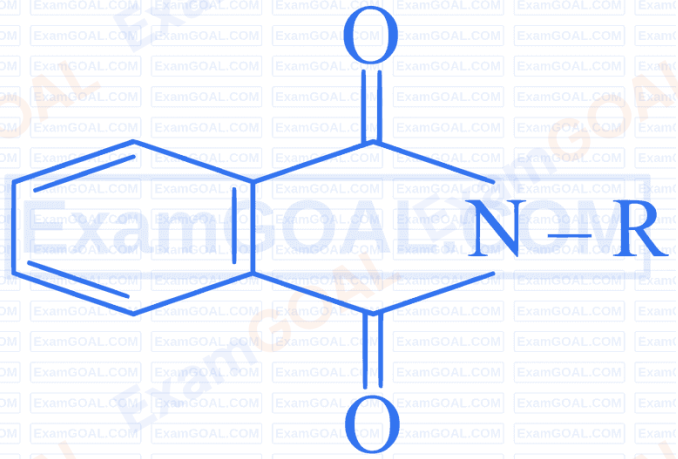
(D) Both Statement I and Statement II are true

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Q20. MCQ (Single Correct Answer)

Open solution

Match the Compounds (List - I) with the appropriate Catalyst/ Reagents (List - II) for their reduction into corresponding amines.

	List - I (Compounds)		List - II (Catalyst/Reagents)
(A)		(I)	$NaOH$ (aqueous)
(B)		(II)	H_2/Ni
(C)	$R-C\equiv N$	(III)	$LiAlH_4, H_2O$
(D)		(IV)	Sn, HCl

Choose the correct answer from the options given below :

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(A) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)
(B) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(C) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
(D) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Q21. Numerical

Consider the following cases of standard enthalpy of reaction $\left(\Delta H_r^{\circ} \text{ in } kJmol^{-1}\right)$

$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \Delta H_1^{\circ} = -1550$

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g) \Delta H_2^{\circ} = -393.5$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \Delta H_3^{\circ} = -286$

The magnitude of $\Delta H_{f C_2H_6(g)}^{\circ}$ is _____ $kJmol^{-1}$ (Nearest integer).

Open solution

Q22. Numerical

The complex of Ni^{2+} ion and dimethyl glyoxime contains _____ number of Hydrogen (H) atoms.

Open solution

Q23. Numerical

Niobium (Nb) and ruthenium (Ru) have " x " and " y " number of electrons in their respective 4 d orbitals. The value of $x + y$ is _____.

Open solution

Q24. Numerical

20 mL of 2 M NaOH solution is added to 400 mL of 0.5 M NaOH solution. The final concentration of the solution is _____ $\times 10^{-2} M$. (Nearest integer)

Open solution

Q25. Numerical

The compound with molecular formula C_6H_6, which gives only one monobromo derivative and takes up four moles of hydrogen per mole for complete hydrogenation has _____ π electrons.

Open solution

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let α_θ and β_θ be the distinct roots of $2x^2 + (\cos\theta)x - 1 = 0, \theta \in (0, 2\pi)$. If m and M are the minimum and the maximum values of $\alpha_\theta^4 + \beta_\theta^4$, then $16(M + m)$ equals :

(A) 27 (B) 17 (C) 25 (D) 24

Open solution

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Q27. MCQ (Single Correct Answer)

Let a line pass through two distinct points $P(-2,-1,3)$ and Q, and be parallel to the vector $3\hat{i}+2\hat{j}+2\hat{k}$. If the distance of the point Q from the point $\mathrm{R}(1,3,3)$ is 5 , then the square of the area of $\triangle PQR$ is equal to :

Open solution

(A) 148 (B) 144 (C) 136 (D) 140

Q28. MCQ (Single Correct Answer)

If the system of linear equations :

$x+y+2z=6$

$2x+3y+\mathrm{a}z=\mathrm{a}+1$

$-x-3y+\mathrm{b}z=2\,\mathrm{b}$

where $a, b \in \mathbf{R}$, has infinitely many solutions, then $7a+3b$ is equal to :

Open solution

(A) 12 (B) 9 (C) 22 (D) 16

Q29. MCQ (Single Correct Answer)

The sum of all values of $\theta \in [0,2\pi]$ satisfying $2\sin^2\theta=\cos 2\theta$ and $2\cos^2\theta=3\sin\theta$ is

Open solution

(A) π (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{2}$ (D) 4π

Q30. MCQ (Single Correct Answer)

Let α, β, γ and δ be the coefficients of x^7, x^5, x^3 and x respectively in the expansion of $\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5, x>1$. If u and v satisfy the equations

Open solution

$\alpha u+\beta v=18,$

$\gamma u+\delta v=20,$

then $\mathrm{u}+\mathrm{v}$ equals :

(A) 4 (B) 3 (C) 5 (D) 8

Q31. MCQ (Single Correct Answer)

The perpendicular distance, of the line $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z+3}{2}$ from the point $\mathrm{P}(2,-10,1)$, is :

Open solution

(A) 6 (B) $4\sqrt{3}$ (C) $3\sqrt{5}$ (D) $5\sqrt{2}$

Q32. MCQ (Single Correct Answer)

If $x=f(y)$ is the solution of the differential equation $\left(1+y^2\right)+\left(x-2\mathrm{e}^{\tan^{-1}y}\right)\frac{\mathrm{d}y}{\mathrm{d}x}=0, y \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ with $f(0)=1$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

Open solution

(A) $\mathrm{e}^{\pi/4}$ (B) $e^{\pi/12}$ (C) $\mathrm{e}^{\pi/6}$ (D) $e^{\pi/3}$

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Q33. MCQ (Single Correct Answer)

If A and B are two events such that $P(A \cap B) = 0.1$, and $P(A \mid B)$ and $P(B \mid A)$ are the roots of the equation $12x^2 - 7x + 1 = 0$, then the value of $\frac{P(\bar{A} \cup \bar{B})}{P(\bar{A} \cap \bar{B})}$ is :

(A) $\frac{4}{3}$ (B) $\frac{7}{4}$ (C) $\frac{9}{4}$ (D) $\frac{5}{3}$

Open solution

Q34. MCQ (Single Correct Answer)

For a 3×3 matrix M, let trace (M) denote the sum of all the diagonal elements of M. Let A be a 3×3 matrix such that $|A| = \frac{1}{2}$ and trace $(A) = 3$. If $B = \text{adj}(\text{adj}(2A))$, then the value of $|B|+$ trace (B) equals :

(A) 56 (B) 132 (C) 174 (D) 280

Open solution

Q35. MCQ (Single Correct Answer)

If $\int e^x \left(\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \frac{\sin^{-1} x}{(1-x^2)^{3/2}} + \frac{x}{1-x^2}\right) dx = g(x) + C$, where C is the constant of integration, then $g\left(\frac{1}{2}\right)$ equals :

(A) $\frac{\pi}{6}\sqrt{\frac{e}{3}}$ (B) $\frac{\pi}{6}\sqrt{\frac{e}{2}}$

(C) $\frac{\pi}{4}\sqrt{\frac{e}{3}}$ (D) $\frac{\pi}{4}\sqrt{\frac{e}{2}}$

Open solution

Q36. MCQ (Single Correct Answer)

Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{3}$. If $\lambda\vec{a} + 2\vec{b}$ and $3\vec{a} - \lambda\vec{b}$ are perpendicular to each other, then the number of values of λ in $[-1, 3]$ is :

(A) 1 (B) 3 (C) 2 (D) 0

Open solution

Q37. MCQ (Single Correct Answer)

The area of the region enclosed by the curves $y = x^2 - 4x + 4$ and $y^2 = 16 - 8x$ is :

(A) $\frac{8}{3}$ (B) 5 (C) 8 (D) $\frac{4}{3}$

Open solution

Q38. MCQ (Single Correct Answer)

In a group of 3 girls and 4 boys, there are two boys B_1 and B_2. The number of ways, in which these girls and boys can stand in a queue such that all the girls stand together, all the boys stand together, but B_1 and B_2 are not adjacent to each other, is :

(A) 120 (B) 96 (C) 72 (D) 144

Open solution

Q39. MCQ (Single Correct Answer)

If $\lim\limits_{x \to \infty} \left(\left(\frac{e}{1-e}\right)\left(\frac{1}{e} - \frac{x}{1+x}\right)\right)^x = \alpha$, then the value of $\frac{\log_e \alpha}{1+\log_e \alpha}$ equals :

(A) e^{-2} (B) e^2 (C) e (D) e^{-1}

Open solution

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Q40. MCQ (Single Correct Answer)

Let $f(x)=\int_0^{x^2}\frac{t^2-8t+15}{e^t}dt, x\in \mathbf{R}$. Then the numbers of local maximum and local minimum points of f, respectively, are :

(A) 3 and 2 (B) 2 and 2 (C) 2 and 3 (D) 1 and 3

Open solution

Q41. MCQ (Single Correct Answer)

Let $A=\{1,2,3,4\}$ and $B=\{1,4,9,16\}$. Then the number of many-one functions $f: A\to B$ such that $1\in f(A)$ is equal to :

(A) 151 (B) 139 (C) 163 (D) 127

Open solution

Q42. MCQ (Single Correct Answer)

Let the curve $z(1+i)+\bar{z}(1-i)=4, z\in C$, divide the region $|z-3|\le 1$ into two parts of areas α and β. Then $|\alpha-\beta|$ equals :

(A) $1+\frac{\pi}{3}$ (B) $1+\frac{\pi}{6}$ (C) $1+\frac{\pi}{2}$ (D) $1+\frac{\pi}{4}$

Open solution

Q43. MCQ (Single Correct Answer)

Suppose that the number of terms in an A.P. is $2k, k\in N$. If the sum of all odd terms of the A.P. is 40 , the sum of all even terms is 55 and the last term of the A.P. exceeds the first term by 27 , then k is equal to:

(A) 8 (B) 6 (C) 4 (D) 5

Open solution

Q44. MCQ (Single Correct Answer)

Let $P(4,4\sqrt{3})$ be a point on the parabola $y^2=4ax$ and PQ be a focal chord of the parabola. If M and N are the foot of perpendiculars drawn from P and Q respectively on the directrix of the parabola, then the area of the quadrilateral PQMN is equal to :

(A) $\frac{34\sqrt{3}}{3}$ (B) $\frac{343\sqrt{3}}{8}$ (C) $17\sqrt{3}$ (D) $\frac{263\sqrt{3}}{8}$

Open solution

Q45. MCQ (Single Correct Answer)

Let $E: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$ and $H: \frac{x^2}{A^2}-\frac{y^2}{B^2}=1$. Let the distance between the foci of E and the foci of H be $2\sqrt{3}$. If $a-A=2$, and the ratio of the eccentricities of E and H is $\frac{1}{3}$, then the sum of the lengths of their latus rectums is equal to :

(A) 10 (B) 7 (C) 9 (D) 8

Open solution

Q46. Numerical

Let $A(6,8), B(10\cos\alpha, -10\sin\alpha)$ and $C(-10\sin\alpha, 10\cos\alpha)$, be the vertices of a triangle. If $L(a,9)$ and $G(h,k)$ be its orthocenter and centroid respectively, then $(5a-3h+6k+100\sin 2\alpha)$ is equal to _____.

Open solution

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Q47. Numerical

Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}, -1 < x < 1$ such that $f(0) = 0$. If $6\int_{-1/2}^{1/2} f(x)dx = 2\pi - \alpha$ then α^2 is equal to _____ .

Open solution

Q48. Numerical

If $\sum_{r=1}^{30} \frac{r^2\left({}^{30}C_r\right)^2}{{}^{30}C_{r-1}} = \alpha \times 2^{29}$, then α is equal to _____.

Open solution

Q49. Numerical

Let the distance between two parallel lines be 5 units and a point P lie between the lines at a unit distance from one of them. An equilateral triangle PQR is formed such that Q lies on one of the parallel lines, while R lies on the other. Then $(QR)^2$ is equal to _____.

Open solution

Q50. Numerical

Let $A = \{1, 2, 3\}$. The number of relations on A, containing $(1, 2)$ and $(2, 3)$, which are reflexive and transitive but not symmetric, is _____.

Open solution

Physics 25 questions

Q51. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : A simple pendulum is taken to a planet of mass and radius, 4 times and 2 times, respectively, that of the Earth. The time period of the pendulum remains same on earth and the planet.

Reason (R) : The mass of the pendulum remains unchanged at Earth and the other planet.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) (A) is false but (R) is true

(C) (A) is true but (R) is false

(D) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Open solution

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Q52. MCQ (Single Correct Answer)

Open solution

Given are statements for certain thermodynamic variables,

(A) Internal energy, volume (V) and mass (M) are extensive variables.

(B) Pressure (P), temperature (T) and density (ρ) are intensive variables.

(C) Volume (V), temperature (T) and density (ρ) are intensive variables.

(D) Mass (M), temperature (T) and internal energy are extensive variables.

Choose the correct answer from the options given below :

(A) (C) and (D) Only **(B)** (A) and (B) Only **(C)** (D) and (A) Only **(D)** (B) and (C) Only

Q53. MCQ (Single Correct Answer)

Open solution

The torque due to the force $(2\hat{i}+\hat{j}+2\hat{k})$ about the origin, acting on a particle whose position vector is $(\hat{i}+\hat{j}+\hat{k})$, would be

(A) $\hat{j}+\hat{k}$ **(B)** $\hat{i}-\hat{k}$ **(C)** $\hat{i}-\hat{j}+\hat{k}$ **(D)** $\hat{i}+\hat{k}$

Q54. MCQ (Single Correct Answer)

Open solution

Which one of the following is the correct dimensional formula for the capacitance in F ? $\mathrm{M}, \mathrm{L}, \mathrm{T}$ and C stand for unit of mass, length, time and charge respectively.

(A) $[\mathrm{F}]=\left[\mathrm{CM}^{-1} \mathrm{~L}^{-2} \mathrm{~T}^{2}\right]$

(B) $[\mathrm{F}]=\left[\mathrm{C}^{2} \mathrm{M}^{-2} \mathrm{~L}^{2} \mathrm{~T}^{2}\right]$

(C) $[\mathrm{F}]=\left[\mathrm{C}^{2} \mathrm{M}^{-1} \mathrm{~L}^{-2} \mathrm{~T}^{2}\right]$

(D) $[\mathrm{F}]=\left[\mathrm{CM}^{-2} \mathrm{~L}^{-2} \mathrm{~T}^{-2}\right]$

Q55. MCQ (Single Correct Answer)

Open solution

The maximum percentage error in the measurement of density of a wire is

[Given, mass of wire $=(0.60 \pm 0.003) \mathrm{g}$

radius of wire $=(0.50 \pm 0.01) \mathrm{cm}$

length of wire $=(10.00 \pm 0.05) \mathrm{cm}]$

(A) 7 **(B)** 8 **(C)** 5 **(D)** 4

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Q56. MCQ (Single Correct Answer)

Open solution

An electron projected perpendicular to a uniform magnetic field B moves in a circle. If Bohr's quantization is applicable, then the radius of the electronic orbit in the first excited state is :

(A) $\sqrt{\frac{h}{\pi eB}}$

(B) $\sqrt{\frac{4h}{\pi eB}}$

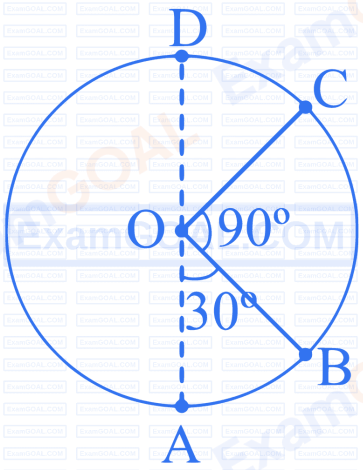
(C) $\sqrt{\frac{\mathrm{h}}{2\pi \mathrm{eB}}}$

(D) $\sqrt{\frac{2h}{\pi eB}}$

Q57. MCQ (Single Correct Answer)

Open solution

A body of mass 100 g is moving in circular path of radius 2 m on vertical plane as shown in figure. The velocity of the body at point A is $10\ \mathrm{m/s}$. The ratio of its kinetic energies at point B and C is :



(Take acceleration due to gravity as $10\ \mathrm{m/s^2}$)

(A) $\frac{3-\sqrt{2}}{2}$

(B) $\frac{2+\sqrt{3}}{3}$

(C) $\frac{2+\sqrt{2}}{3}$

(D) $\frac{3+\sqrt{3}}{2}$

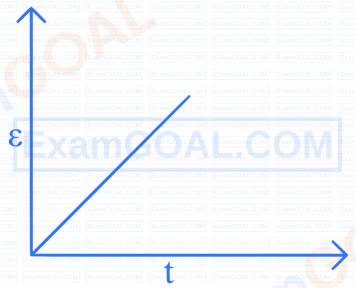
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Q58. MCQ (Single Correct Answer)

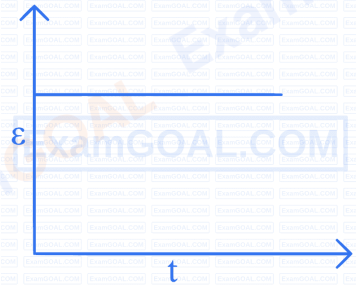
Open solution

A rectangular metallic loop is moving out of a uniform magnetic field region to a field free region with a constant speed. When the loop is partially inside the magnetic field, the plot of magnitude of induced emf (ε) with time (t) is given by

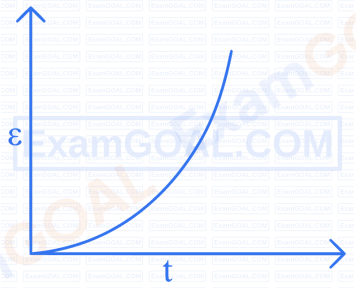
(A)



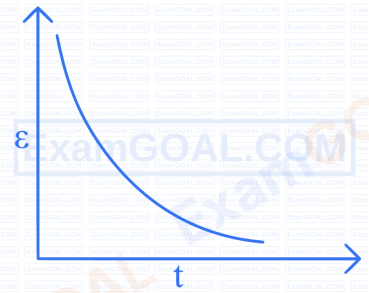
(B)



(C)

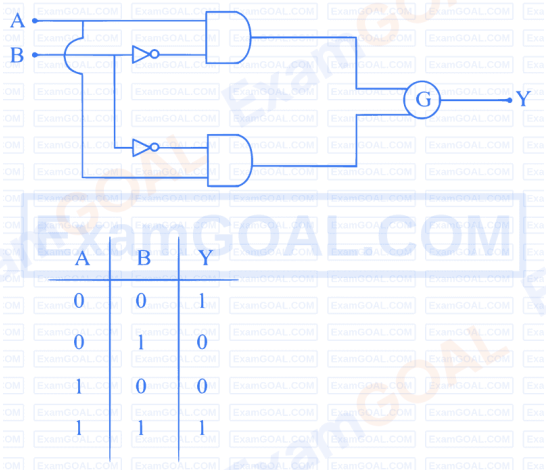


(D)



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Q59. MCQ (Single Correct Answer)



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

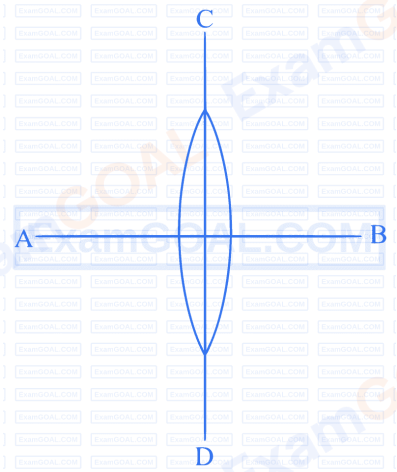
Open solution

To obtain the given truth table, following logic gate should be placed at G :

(A) NOR Gate **(B)** OR Gate **(C)** AND Gate **(D)** NAND Gate

Q60. MCQ (Single Correct Answer)

A symmetric thin biconvex lens is cut into four equal parts by two planes AB and CD as shown in figure. If the power of original lens is 4D then the power of a part of the divided lens is



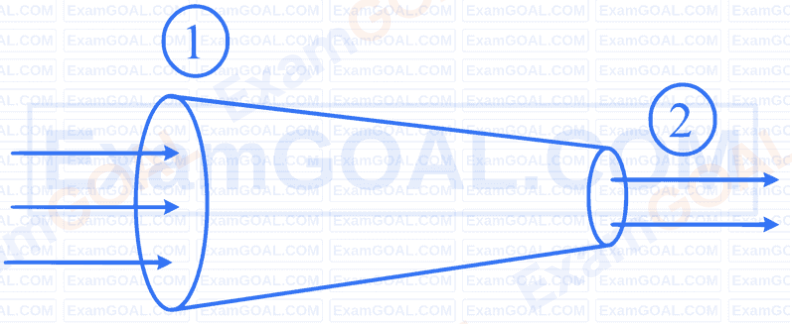
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(A) 2D **(B)** 8D **(C)** 4D **(D)** D

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Q61. MCQ (Single Correct Answer)

Open solution



A tube of length L is shown in the figure. The radius of cross section at the point (1) is 2 cm and at the point (2) is 1 cm , respectively. If the velocity of water entering at point (1) is $2\ \mathrm{m/s}$, then velocity of water leaving the point (2) will be

(A) 4 m/s (B) 6 m/s (C) 2 m/s (D) 8 m/s

Q62. MCQ (Single Correct Answer)

Open solution

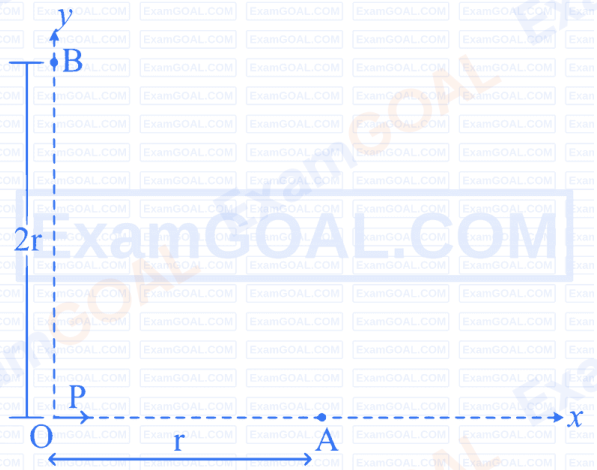
A series LCR circuit is connected to an alternating source of emf E. The current amplitude at resonant frequency is I_0. If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be

(A) $\frac{I_0}{2}$ (B) $\frac{I_0}{\sqrt{2}}$ (C) $2I_0$ (D) I_0

Q63. MCQ (Single Correct Answer)

Open solution

For a short dipole placed at origin O , the dipole moment P is along x-axis, as shown in the figure. If the electric potential and electric field at A are V_0 and E_0, respectively, then the correct combination of the electric potential and electric field, respectively, at point B on the y-axis is given by



(A) $\frac{V_0}{2}$ and $\frac{E_0}{16}$ (B) zero and $\frac{E_0}{8}$

(C) V_0 and $\frac{E_0}{4}$ (D) zero and $\frac{E_0}{16}$

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Q64. MCQ (Single Correct Answer)

A transparent film of refractive index, 2.0 is coated on a glass slab of refractive index, 1.45. What is the minimum thickness of transparent film to be coated for the maximum transmission of Green light of wavelength 550 nm . [Assume that the light is incident nearly perpendicular to the glass surface.]

Open solution

(A) 68.7 nm (B) 137.5 nm (C) 94.8 nm (D) 275 nm

Q65. MCQ (Single Correct Answer)

A light source of wavelength λ illuminates a metal surface and electrons are ejected with maximum kinetic energy of 2 eV . If the same surface is illuminated by a light source of wavelength $\frac{\lambda}{2}$, then the maximum kinetic energy of ejected electrons will be (The work function of metal is 1 eV)

Open solution

(A) 5 eV (B) 3 eV (C) 2 eV (D) 6 eV

Q66. MCQ (Single Correct Answer)

For a diatomic gas, if $\gamma_1 = \left(\frac{Cp}{Cv}\right)$ for rigid molecules and $\gamma_2 = \left(\frac{Cp}{Cv}\right)$ for another diatomic molecules, but also having vibrational modes. Then, which one of the following options is correct? (Cp and Cv are specific heats of the gas at constant pressure and volume)

Open solution

(A) $\gamma_2 < \gamma_1$ (B) $\gamma_2 > \gamma_1$ (C) $\gamma_2 = \gamma_1$ (D) $2\gamma_2 = \gamma_1$

Q67. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In Young's double slit experiment, the fringes produced by red light are closer as compared to those produced by blue light.

Reason (R) : The fringe width is directly proportional to the wavelength of light.

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true but $(\mathbf{R})$ is NOT the correct explanation of (A)

Q68. MCQ (Single Correct Answer)

A ball of mass 100 g is projected with velocity $20\ \mathrm{m/s}$ at 60° with horizontal. The decrease in kinetic energy of the ball during the motion from point of projection to highest point is

Open solution

(A) 20 J (B) 5 J (C) 15 J (D) zero

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Q69. MCQ (Single Correct Answer)

A force $\overrightarrow{F} = 2\hat{i} + b\hat{j} + \hat{k}$ is applied on a particle and it undergoes a displacement $\hat{i} - 2\hat{j} - \hat{k}$. What will be the value of b, if work done on the particle is zero.

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 0 (D) 2

Open solution

Q70. MCQ (Single Correct Answer)

A small rigid spherical ball of mass M is dropped in a long vertical tube containing glycerine. The velocity of the ball becomes constant after some time. If the density of glycerine is half of the density of the ball, then the viscous force acting on the ball will be (consider g as acceleration due to gravity)

(A) 2 Mg (B) $\frac{3}{2}$Mg (C) $\frac{Mg}{2}$ (D) Mg

Open solution

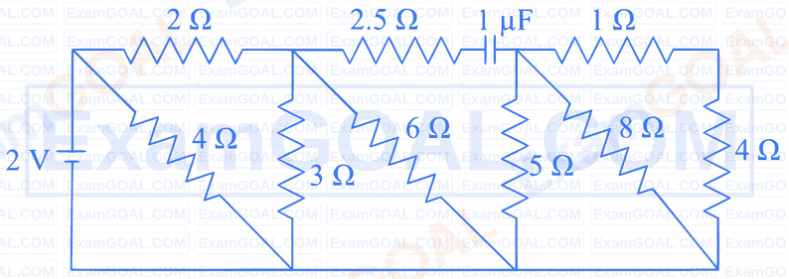
Q71. Numerical

A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of $2 \times 10^5\ ms^{-1}$. When the electric field is switched off, the proton moves along a circular path of radius 2 cm . The magnitude of electric field is $x \times 10^4\ N/C$. The value of x is _____. Take the mass of the proton $= 1.6 \times 10^{-27}$ kg.

Open solution

Q72. Numerical

The net current flowing in the given circuit is _____ A.



Open solution

Q73. Numerical

A parallel plate capacitor of area $A = 16\ cm^2$ and separation between the plates 10 cm , is charged by a DC current. Consider a hypothetical plane surface of area $A_0 = 3.2\ cm^2$ inside the capacitor and parallel to the plates. At an instant, the current through the circuit is 6A. At the same instant the displacement current through A_0 is _____ mA .

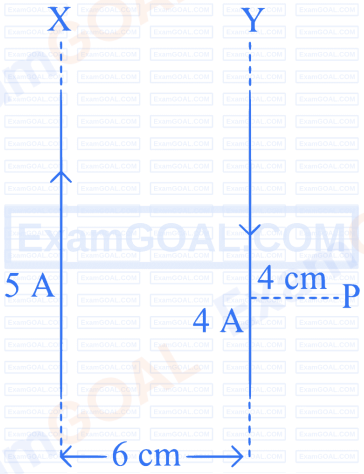
Open solution

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Q74. Numerical

Two long parallel wires X and Y, separated by a distance of 6 cm , carry currents of 5 A and 4A, respectively, in opposite directions as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is $x \times 10^{-5}$ T. The value of x is _____ . Take permeability of free space as $\mu_0 = 4\pi \times 10^{-7}$ SI units.

Open solution



Q75. Numerical

A tube of length 1 m is filled completely with an ideal liquid of mass 2 M , and closed at both ends. The tube is rotated uniformly in horizontal plane about one of its ends. If the force exerted by the liquid at the other end is F then angular velocity of the tube is $\sqrt{\frac{F}{\alpha M}}$ in SI unit. The value of α is _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **C**	4. **A**	5. **C**
6. **A**	7. **B**	8. **D**	9. **C**	10. **A**
11. **C**	12. **C**	13. **A**	14. **C**	15. **D**
16. **C**	17. **D**	18. **A**	19. **A**	20. **B**
21. **95**	22. **14**	23. **11**	24. **57**	25. **8 or 6**

Mathematics

26. **C**	27. **C**	28. **D**	29. **A**	30. **C**
31. **C**	32. **C**	33. **C**	34. **D**	35. **A**
36. **D**	37. **A**	38. **D**	39. **C**	40. **C**
41. **A**	42. **C**	43. **D**	44. **B**	45. **D**
46. **145**	47. **27**	48. **465**	49. **28**	50. **3**

Physics

51. **D**	52. **B**	53. **B**	54. **C**	55. **C**
56. **A**	57. **D**	58. **B**	59. *Scan QR*	60. **A**
61. **D**	62. **A**	63. **D**	64. **B**	65. **A**
66. **A**	67. **C**	68. **C**	69. **B**	70. **C**
71. **2**	72. **1**	73. **1200**	74. **1**	75. **1**

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